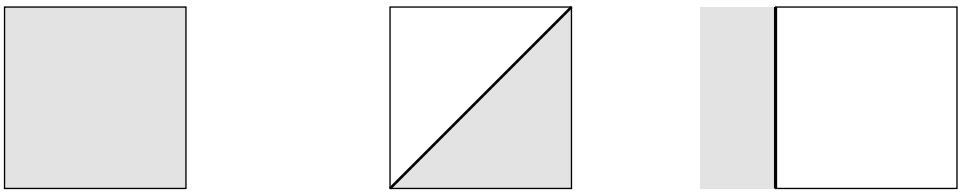
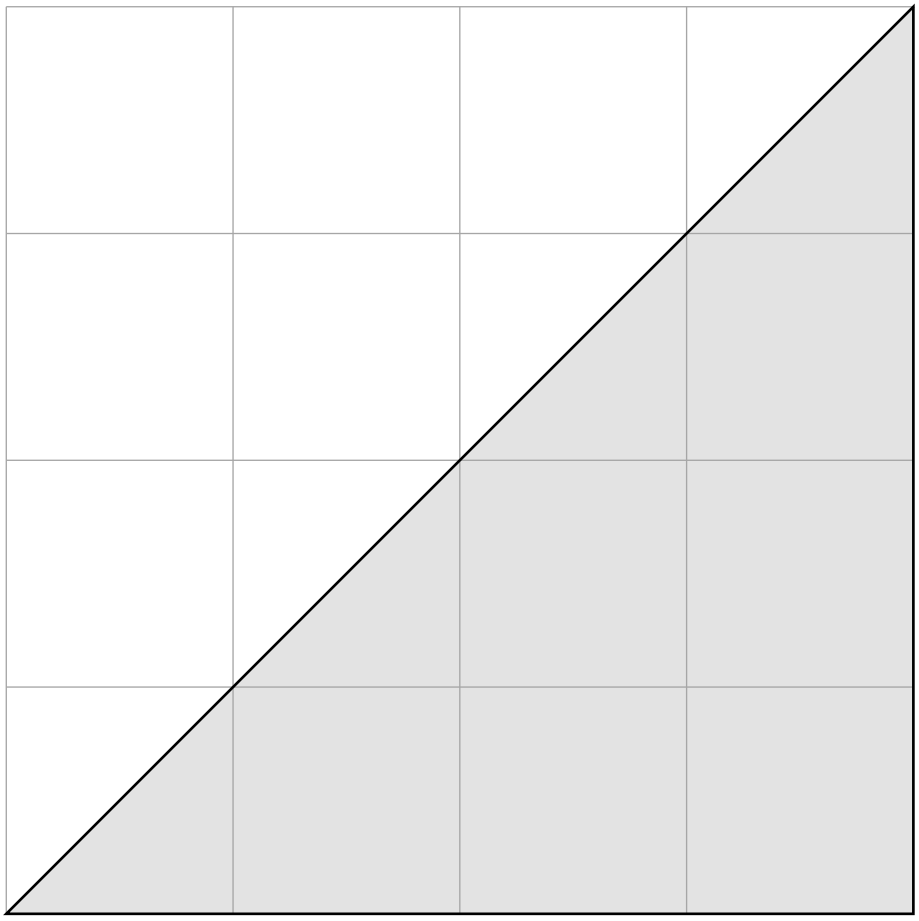
Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.



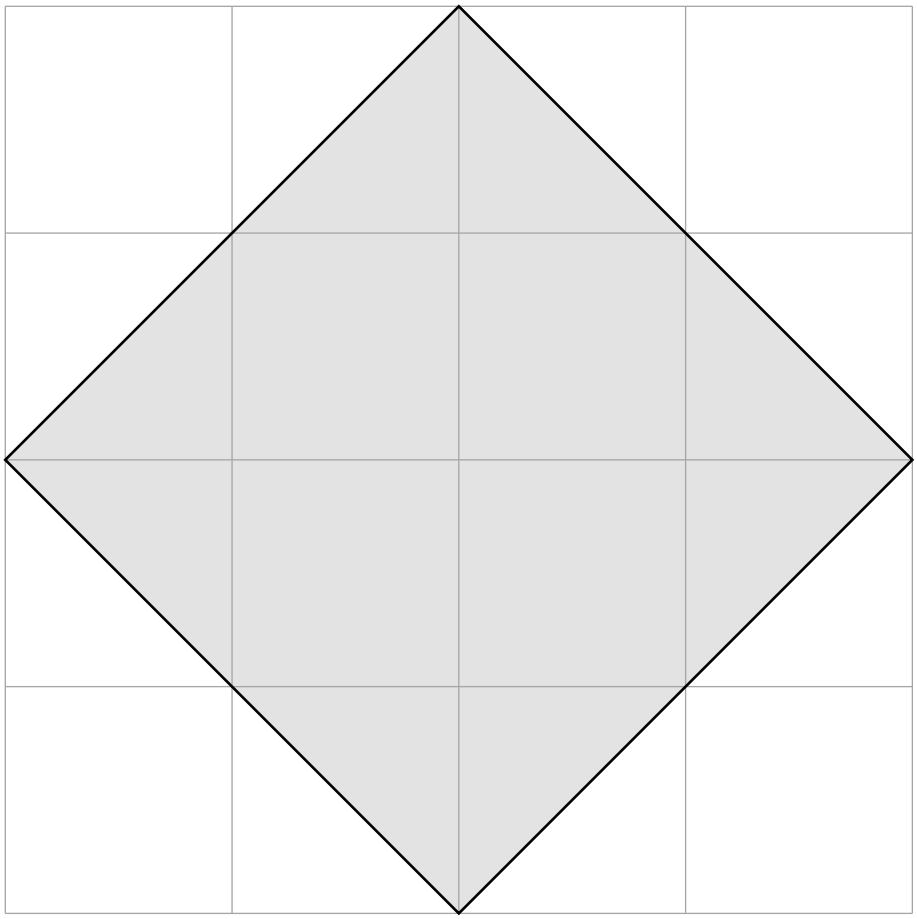
whole partial outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.



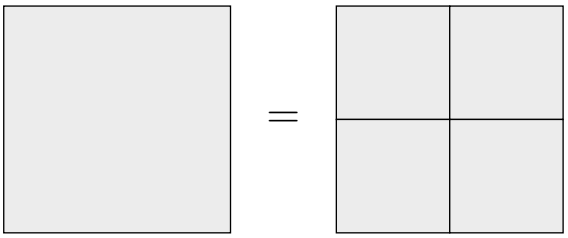
Inside: _____ big squares Cover: _____ big squares

Problem 2: Trap the diamond's area between its whole squares and its cover. Can you be certain its area is more than 3 big squares? Less than 13?



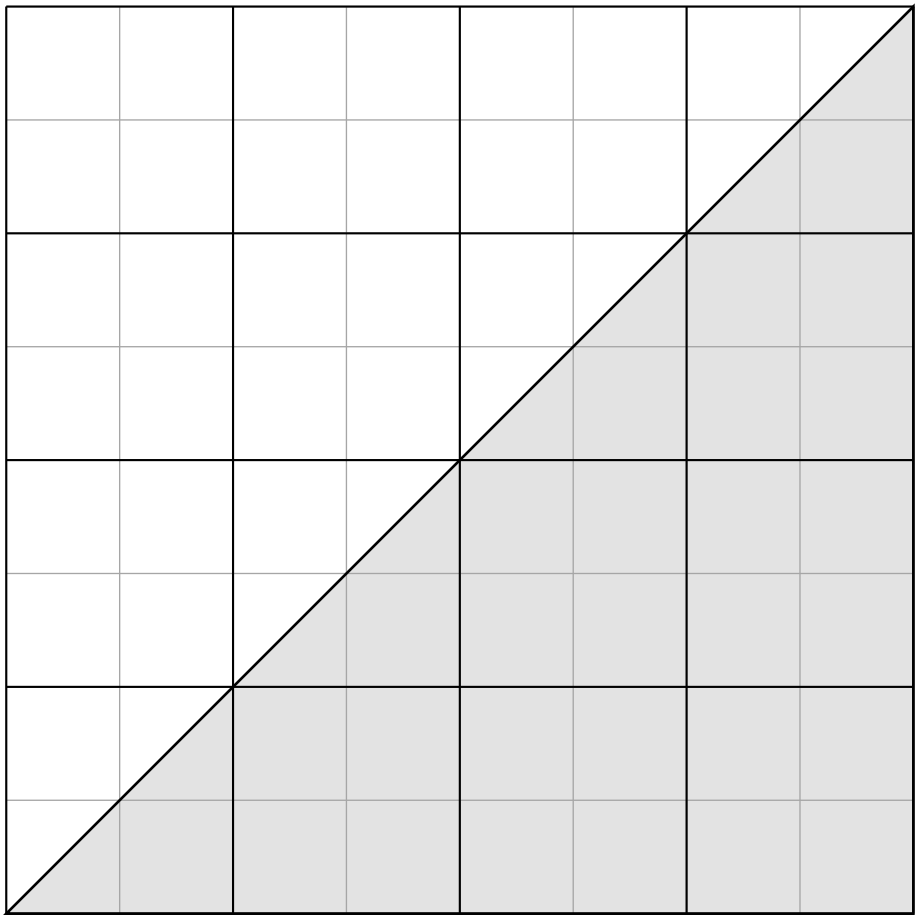
Inside: _____ big-square units

Cover: _____ big-square units



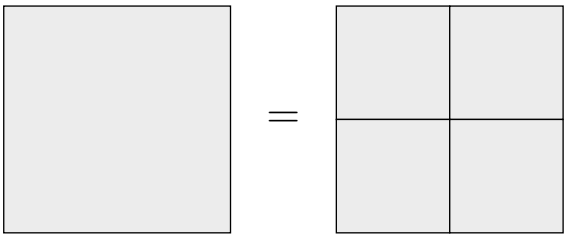
1 big square
= 4 small squares

Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.



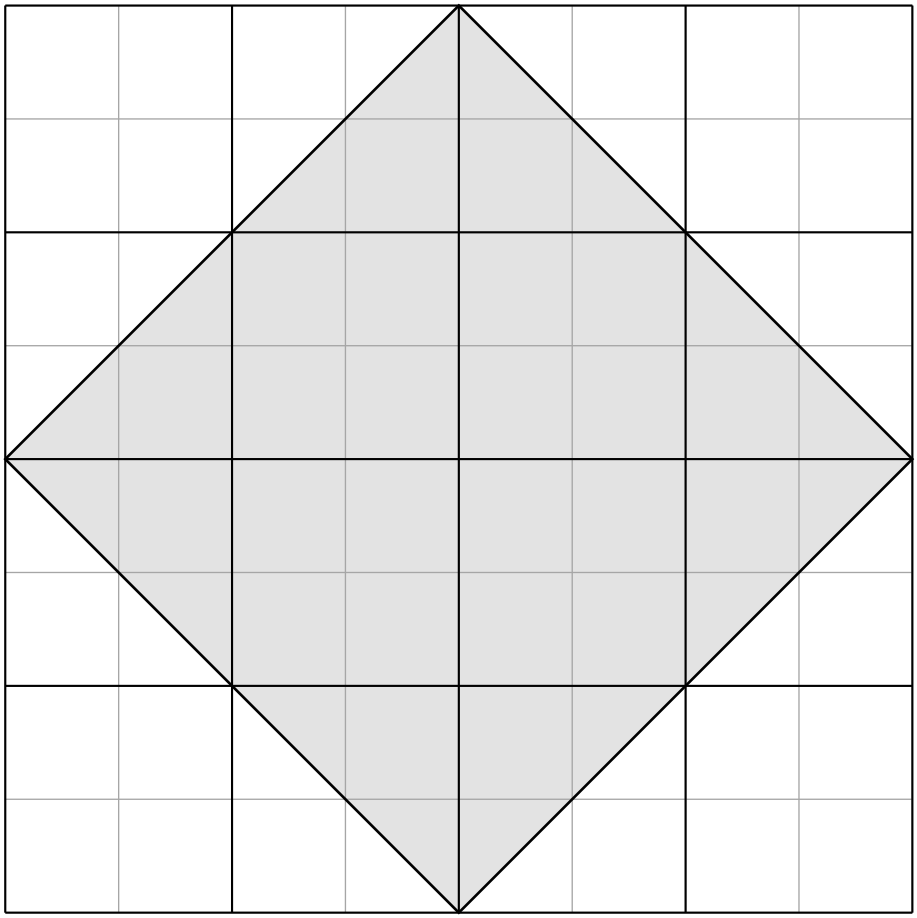
Inside: _____ big-square units

Cover: _____ big-square units



1 big square
= 4 small squares

Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.



Inside: _____ big-square units

Cover: _____ big-square units